



SAHAYAK

Offline. Vernacular. On your side.

A scam shield and benefits guide for people new to digital money: at the CSC counter and on their own phone, with no internet, in Hindi and English, by voice or touch. Built and measured.

Ideas for India 2026 · Inclusive Innovation for Bharat · Financial Inclusion

Roshan Raj · Ishmiit Singh

Banked, then left to be robbed

Hundreds of millions of first-time account holders now get OTPs, UPI requests and scheme messages they were never taught to read, and miss benefits they are owed.

₹22,495 crore

lost to financial cyber fraud in 2025, across 24 lakh complaints

₹4,005 crore

lost by senior citizens in 2025 (1,03,488 complaints)

1 in 5

families with a UPI user hit by UPI fraud in the last 3 years; 51% never reported it

55%

of people aged 60+ know the old-age pension exists; 29% of poor elders receive it

And most safety tools assume the internet, English, an app account and the cloud, and a person who already knows what a scam looks like: what the people most at risk are least likely to have, or trust.

At the counter, and on the phone at home

One small computer at a CSC or bank-agent counter; phones join its Wi-Fi, which has no internet. A phone that has opened Sahayak's secure web address once (the public site today; the node's own after a one-time HTTPS setup) runs the same checks with no node at all.

"Is this a scam?"

Paste, speak, photograph or describe a message, a call or a UPI QR. A verdict in milliseconds, the three reasons, what to do now, and a ready 1930 complaint.

"What am I owed?"

A spoken interview (median 5 questions) across 12 central schemes, health first: what you can get, the papers, where to go, what to say, and a printed slip.

Then, on their own phone

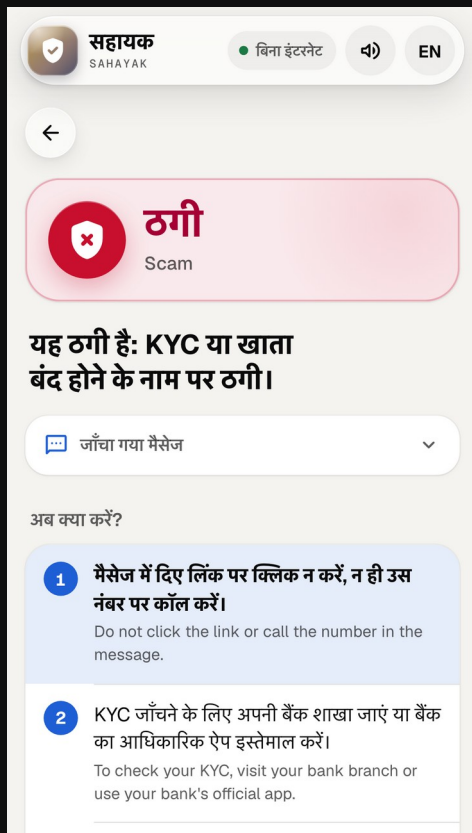
At home, when the call actually comes: the same scam check and benefits interview, on the phone, with no node and no internet. Nothing leaves the phone.



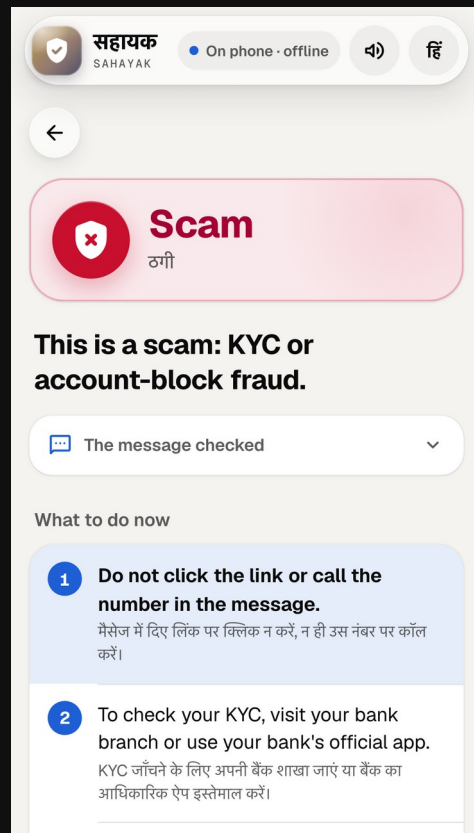
Try it now, on your own phone: roshworldwide.github.io/optum-v2

Check a message, switch on airplane mode, check another: it still works, and gives the node's exact answer.

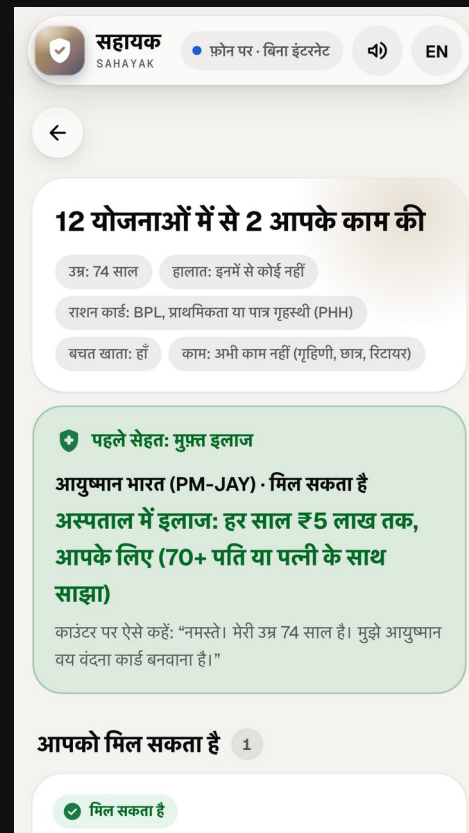
Built, and on a phone today



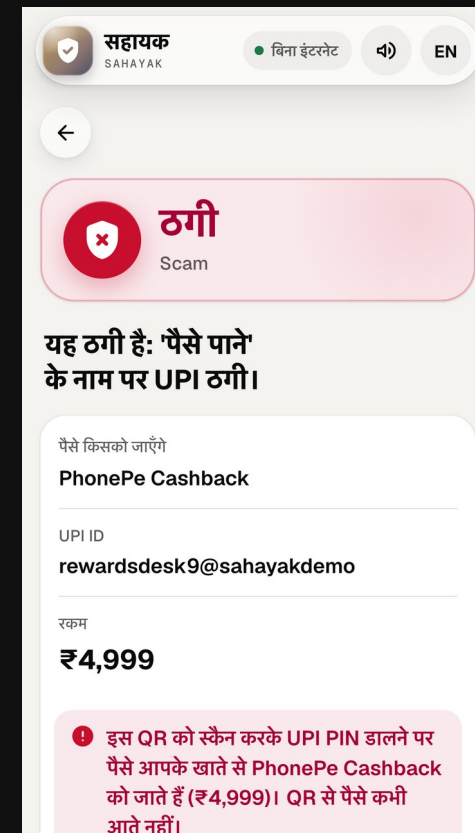
A fake KYC SMS, on the node



Airplane mode: checked on the phone



Age 74: health cover first



"Scan to get cashback" QR

Technology and implementation

- | | | |
|----|-------------------------------|---|
| 01 | Phone app | A web app, no install, Hindi and English, every screen read aloud. After one visit it works offline on the phone. |
| 02 | Fraud-Shield | 64 named signals in a signed content pack, a pattern matcher and a small classifier decide. Vetted templates explain; an optional local model may add a line, only through a safety gate. |
| 03 | Benefits Navigator | Scheme rules as data, each traced to its official page and date; yes / likely / unknown / no logic; asks only what can change the answer. |
| 04 | Offline voice | Speech in and out on the node (Vosk, Piper); amounts spoken as words; "I heard ..., is that right?" before any answer is used. |
| 05 | The node | One mini-PC or laptop at a CSC. Zero-egress monitor, Ed25519-signed packs updated by USB, operator console, captive portal. |
| 06 | Same answer everywhere | The phone runs the same rules from the same signed packs: identical results to the node on 1,841 messages and 7,983 interviews. |

We tested the big-model idea, and chose rules

Round 1 proposed

A 14-billion-parameter language model, offline on a GPU server, deciding whether a message is a scam, with a fine-tuned finance model behind it.

We built

64 named signals, a pattern matcher and a small classifier decide, in about 2 ms on a laptop CPU, and name their reasons. A language model is optional: it may reword the explanation, through a safety gate, and never decides.

Because we measured

A qwen2.5 3b model, asked zero-shot on 137 real published messages, caught 99 of 103 scams (Sahayak 67) but flagged 21 of 34 genuine ones (Sahayak 11), at 2.13 s a message on a GPU. Blind set: 24 of 64 genuine flagged (Sahayak 15).

A warning that fires on 6 in 10 real genuine messages teaches people to ignore warnings, and the model cannot say what made it decide. Where a model earns its place: it caught 14 of 15 blind-test scams in languages Sahayak cannot read yet, so next it becomes a raise-only second opinion there.

What already exists, and what Sahayak adds

Already out there	What it does well	What Sahayak adds
Airtel and Jio network filters	Block spam calls and links at network scale; Airtel reports 68.7% lower fraud losses	Checks the message, QR code or call story in front of the person and says why, in Hindi, offline
Truecaller	Caller ID and spam lists for 339 million users	No account and no upload: nothing leaves the phone; covers UPI QR codes and screenshots
Google on-device scam alerts	On-device call analysis	Runs in any phone's browser; Google's works on Pixel 9+ only (under 1% of phones)
1930, cybercrime.gov.in, Sanchar Saathi	Reporting and blocking after the fraud	Catches it before the payment, and ends every scam verdict in a ready 1930 complaint
myScheme	4,700+ schemes in 15 languages, online, self-serve	Offline, spoken interview with a slip for the CSC; every rule and amount proven against official pages
Haqdarshak	₹19,233 crore in benefits through trained agents	The offline front door and the scam shield; a content partner, not a rival

Measured on real messages, not just our own

69%

of 97 real published scams caught,
first run (keyword blocklist 64%)

89%

of scams caught in a blind test
written by a separate AI model
(blocklist 50%)

92%

of our own test scams caught
(blocklist 51%)

2 ms

per verdict on a laptop CPU

15.4%

Hindi word error, adults reading
aloud

How we know, and what it does not show

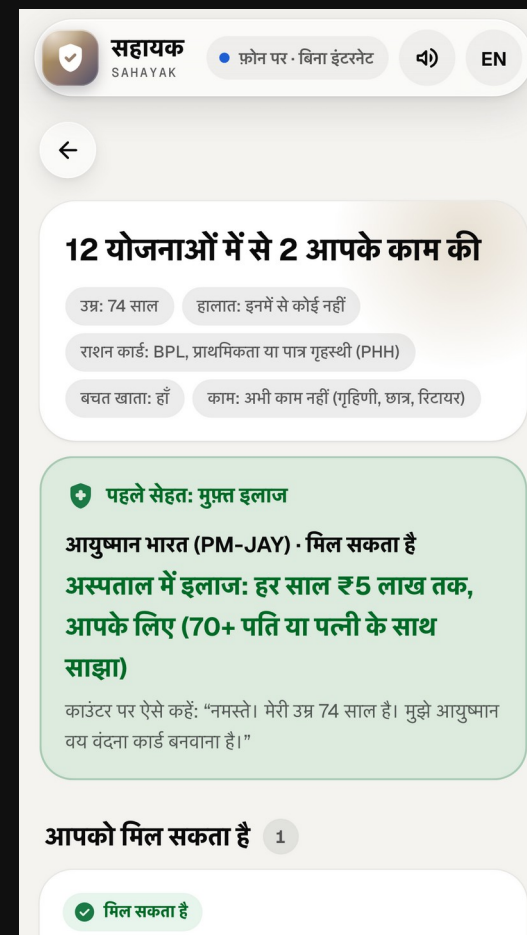
Real messages: 137 that people received, published by the Income Tax portal, PIB Fact Check, courts and fact-checkers, scored once: 67 of 97 scams caught, 11 of 34 genuine flagged (blocklist 17). Fixes then learned from half of them; on the other half, never read, scams caught went from 28 to 31 of 44 and genuine flagged from 6 to 6 of 20. Blind red team, 182 messages: 92 of 103 caught, 14 of 59 hard genuine flagged; other languages get "could not check", never a false green. Our own test split: 34 of 37 (95% Wilson CI 79%–97%). Not yet: messages from people's own phones, and real users.

₹5 lakh of health cover most seniors have not claimed

1.1 crore of ~6 crore

seniors aged 70+ had their Ayushman Vay Vandana card by Feb 2026

- Every benefits result leads with Ayushman Bharat; for anyone 70 or older it is the Vay Vandana card: up to ₹5 lakh a year of hospital care whatever their income (shared with a spouse who is also 70 or more; not for outpatient visits).
- One question, their age, finds them; the slip tells the operator what to make.
- The CSC makes the card at the counter, and is paid ₹20 per first-time card; the citizen pays nothing.
- The scam check flags "pay to get your Ayushman card" messages; the node counts seniors found, every month, with no personal data.



From "account opened" to "citizen safe"

Rupees at risk

The node adds up the money asked for in messages it flags, every month: never called "saved", always shown to the district without personal data.

People found, health first

People found eligible, by scheme, with seniors found for Ayushman Vay Vandana counted on their own; each slip lets the operator finish the form.

Reported in time

From 1 Jan 2027, RBI compensates small losses from stolen OTPs and PINs if reported within 5 days; Sahayak drafts the complaint on the spot. Money sent willingly is not compensated: there, stopping the payment is the only protection.

Pilot target, not a result

One district: about 20 villages and 50 counters (nodes at 20 CSCs, the phone app at 30 bank-agent points), six months, ₹39.2 lakh line by line. Scams flagged, people found eligible and cards made (the operator ticks each), all from the node's counters. A one-morning field test at a CSC (protocol and consent ready) comes first.

Grow by replication; paid for by those who gain

Replication, not reinvention

- 6 lakh functional CSCs, 4.7 lakh of them at gram-panchayat level: the counters already exist.
- A deployment is a node image, signed packs and a one-page operator guide; a new scam or scheme is a pack update by USB.
- Every phone that visits can keep its own copy, so protection spreads beyond the counter at no extra cost.
- No cloud and no per-query fee; open, tested rules any state can audit.

Who pays

₹25,000–35,000

once per node (an N150 mini-PC (8–16 GB) or a refurbished i5 laptop); each check costs nothing.

- Banks: financial-inclusion or CSR budgets for their agent points, as customer education; fewer small-fraud payouts (shared with RBI from 1 Jan 2027) are a bonus, not the case.
- The CSC network: paid about ₹20 per first-time Ayushman card or e-Shram registration Sahayak sends to the counter (how much reaches the operator varies); 86% of rural operators earn under ₹500 a month.
- The State Health Agency and the district: Vay Vandana card drives at the CSCs, funded against a signed monthly count of cards made.

Inclusive innovation on India's own rails

Financial inclusion, made safe

Protects the 59 crore Jan Dhan accounts and the UPI base the nation built, for the people who opened them last.

1930 and RBI's 2027 rules

Every scam verdict ends in the national helpline; fast reporting is what RBI's compensation now requires.

Ayushman Bharat

Finds seniors for the Vay Vandana card and sends them to the counter that makes it.

Digital India, last mile

Hindi and English by voice today, offline; the language data is a pack, so more languages follow.

DPDP Act

Privacy by design: consent first, nothing named, nothing written to disk, deleted by default.

CSC · BC · ASHA

Delivered through the public networks people already trust, with an operator at the counter.

Built in the open; seeking last-mile partners

CSC village-level entrepreneur or bank business correspondent

hosts the node and the field morning

seeking

A rural NGO or SHG federation

invites participants, checks the Hindi and the consent

seeking

A bank branch, regional office or SLBC

BC points for the pilot, real messages for ScamBench v1

seeking

Haqdarshak or myScheme

scheme content beyond 12 central schemes; Sahayak is the offline front door

seeking

Ready for each partner: a one-morning field protocol with consent in Hindi and English, a scoring script, and letters of intent. The code is MIT-licensed and open.

Builders who ship, asking for one district

Roshan Raj

Systems and AI-infrastructure engineer · Manipal Institute of Technology

Ishmiit Singh

Co-builder: product, content and evidence

Together: the node, the scam shield, the scheme navigator, offline voice, the phone engine, the operator console and the benchmarks behind every number in this deck.

The ask

A six-month, single-district pilot (about 20 villages, 50 counters, roughly ₹35–40 lakh) with:

- CSC e-Governance or the State IT department: the counters
- A bank BC network or SLBC: reach, and real messages for ScamBench v1
- NHA or the state health agency: Vay Vandana cards made at the counter
- A rural NGO: field testing and Hindi review

India banked half a billion people.

Sahayak helps keep them safe, and claim what they are owed.

At the counter and on their own phone · offline · in their language · nothing leaves the device · built and measured. Starting with ₹5 lakh of health cover for every senior who does not yet have the card.

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roshworldwide.github.io/optum-v2